Practical 4: PGE petrography and volumetrics calculation for a PGE deposit

Part A

Today we will illustrate the petrography of typical PGE-bearing ultramafic and mafic rocks from layered intrusions. For this, we make use of a number of hand specimens (HS) and thin sections (TS). Please use HS and TS to identify the main rock forming minerals (see listing on next page, and remember your first year classes!), and the fabrics of the rocks, including layering and cumulates.

Hand samples illustrate:
- pegmatite fabric (pegmatite = very coarse-grained igneous rock)
- layered fabric
- massive chromitite

(careful, samples are heavy)

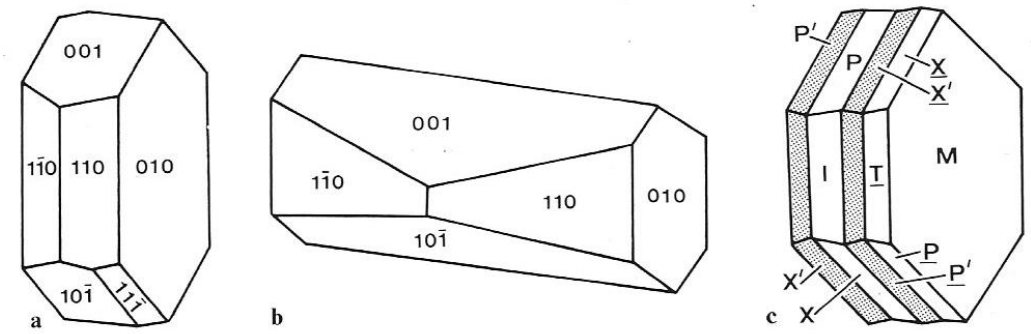
There are several thin sections showing the mineralogy and chromite cumulates.

Also, if you have time either during the practical, or at home, have a look at these online examples:
https://www.virtualmicroscope.org/content/troctolite-chromite-layer
https://www.virtualmicroscope.org/content/o3-magnetite-gabbro
https://www.virtualmicroscope.org/content/merensky-reef-bushveld-complex-elements

Typical crystal morphologies of some of these minerals:

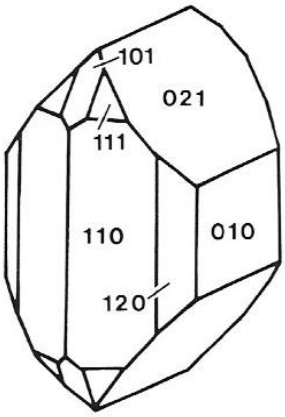
Feldspar:

Transparent minerals, low order grey interference colors, well developed twinning



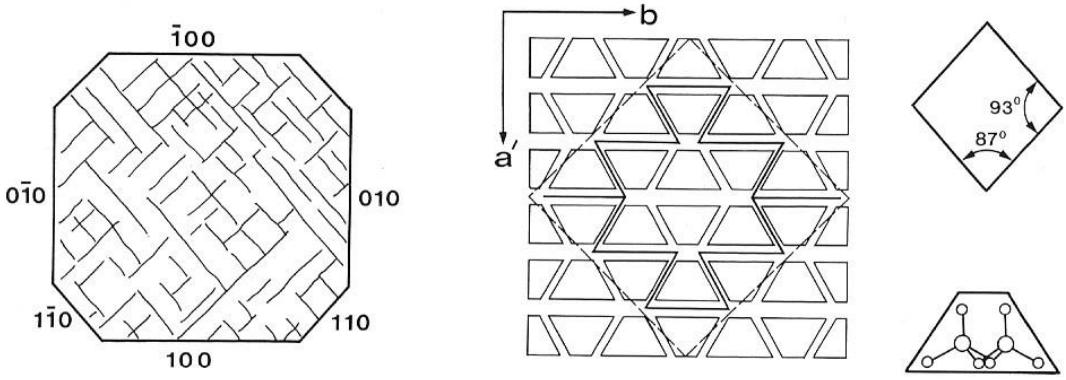
Olivine:

equigranular minerals, high relief, high order interference colors, well developed cleavage



Pyroxene:

high relief minerals with distinct near 90° cleavage



Part B

This calculation illustrates the basic simplicity of maths used in reserve/resource calculations. Note that a number of uncertainties such as ore losses are not considered here. Full evaluations would also take a statistical approach. Data modified from Grobler (2019): https://doi.org/10.1007/s00126-018-0800-x

Data:

Consider a prospect on the northern limb of the Bushveld Complex, exploring for PGEs and Cr. The extract from the geological map, block diagram and cross section on the next 2 pages (Figs. 1-3) show the stratigraphy, dip of strata, main faults, and exploration data (boreholes, cross sections). The prospect area is outlined in the red dashed line. Exploration focuses on several units, including the Tennis Ball Marker (TBM, shown in blue on the map and block diagram).

Although in reality the rocks are folded, for this exercise consider the TBM as a simple sheet of rock with a thickness of 0.8 m.

The reef consists of a pegmatitic pyroxenite with the following modal composition:
70 wt% pyroxene (diopside, $CaMgSi_2O_6$)
25 wt% plagioclase (anorthite, $CaAl_2Si_2O_8$)
2 wt% sulfide (pyrrhotite, $Fe_{1-x}S$)
3 wt% chromite, $FeCr_2O_4$

Questions:

1) What is the total resource of platinum in this prospect?

2) The resource calculated is for total PGEs – assuming 1/3 of this by weight is Pt, and considering the cost of extracting an ounce of Pt is 1000$, check current market prices for Pt to see if mining is economically viable. What parameters can change around economic viability?

3) Calculate the expected temperature in a mine shaft at 1km TVD. A map of geothermal gradients is attached (Figs. 4, 5).

Homework: What is the total resource down to 1 km vertical depth (typical Bushveld mine depth)?

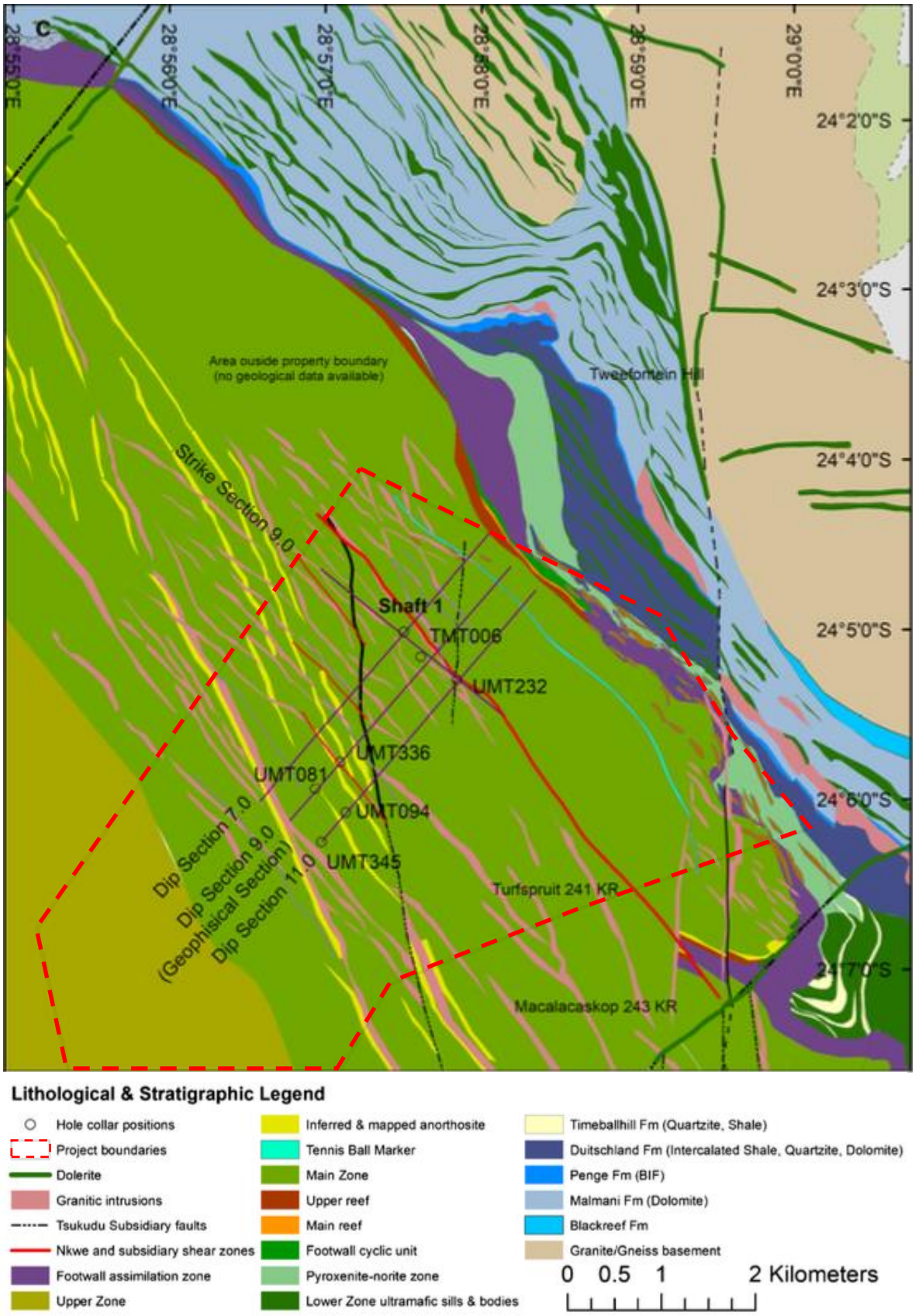


Fig. 1: Geological map (Grobler 2019)

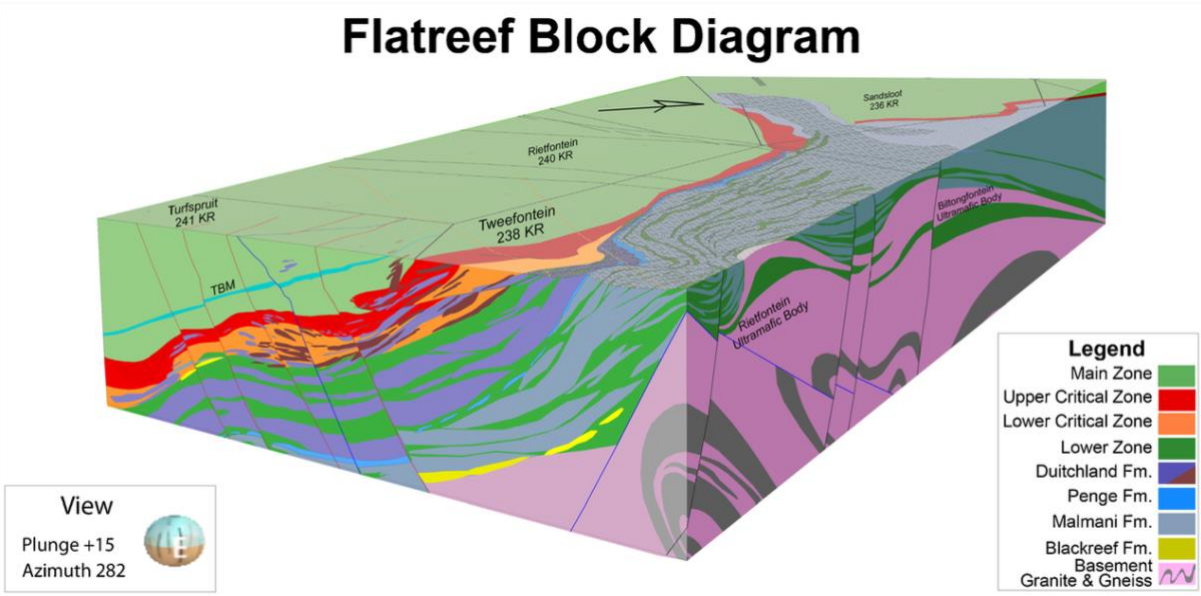


Fig. 2: Block diagram (Grobler 2019)

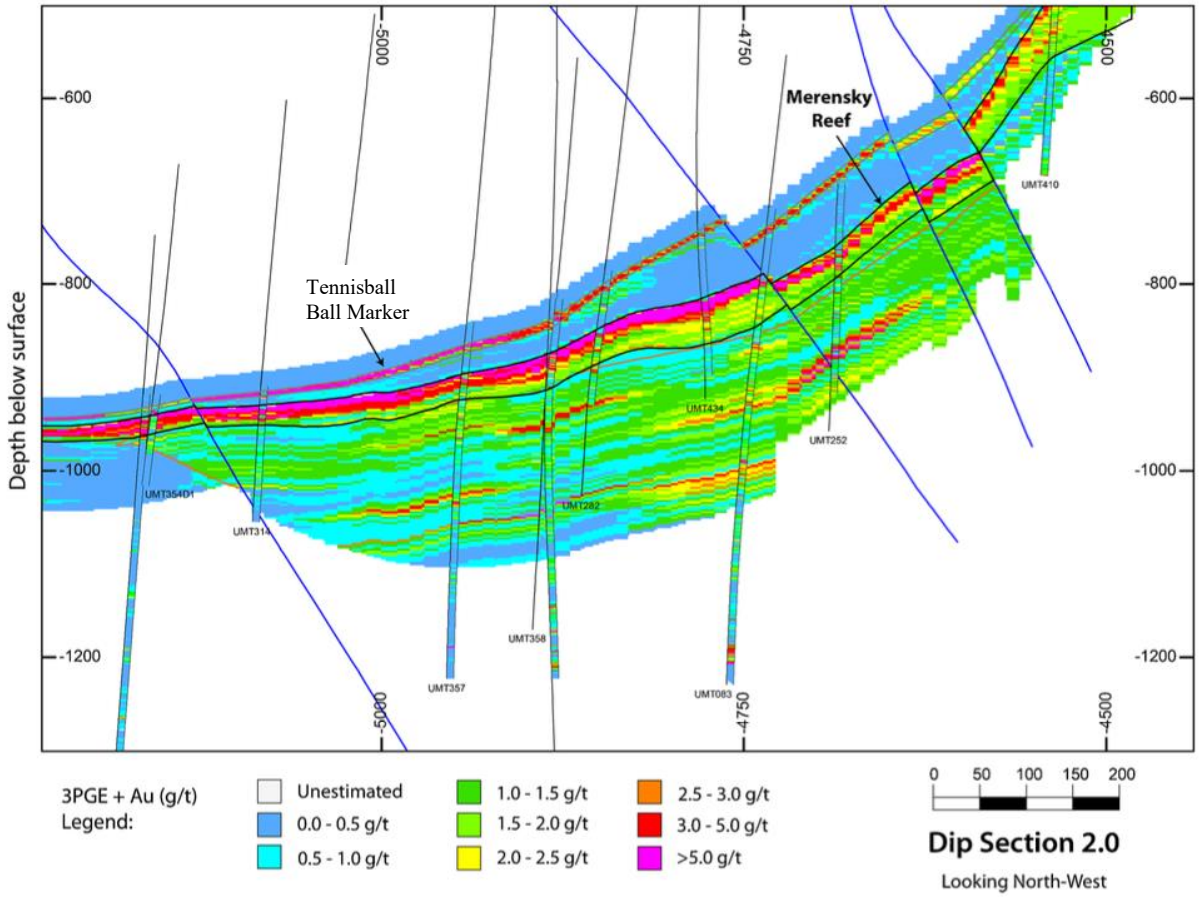


Fig. 3: Cross section of part of the prospect with borehole-based grades (Grobler 2019)

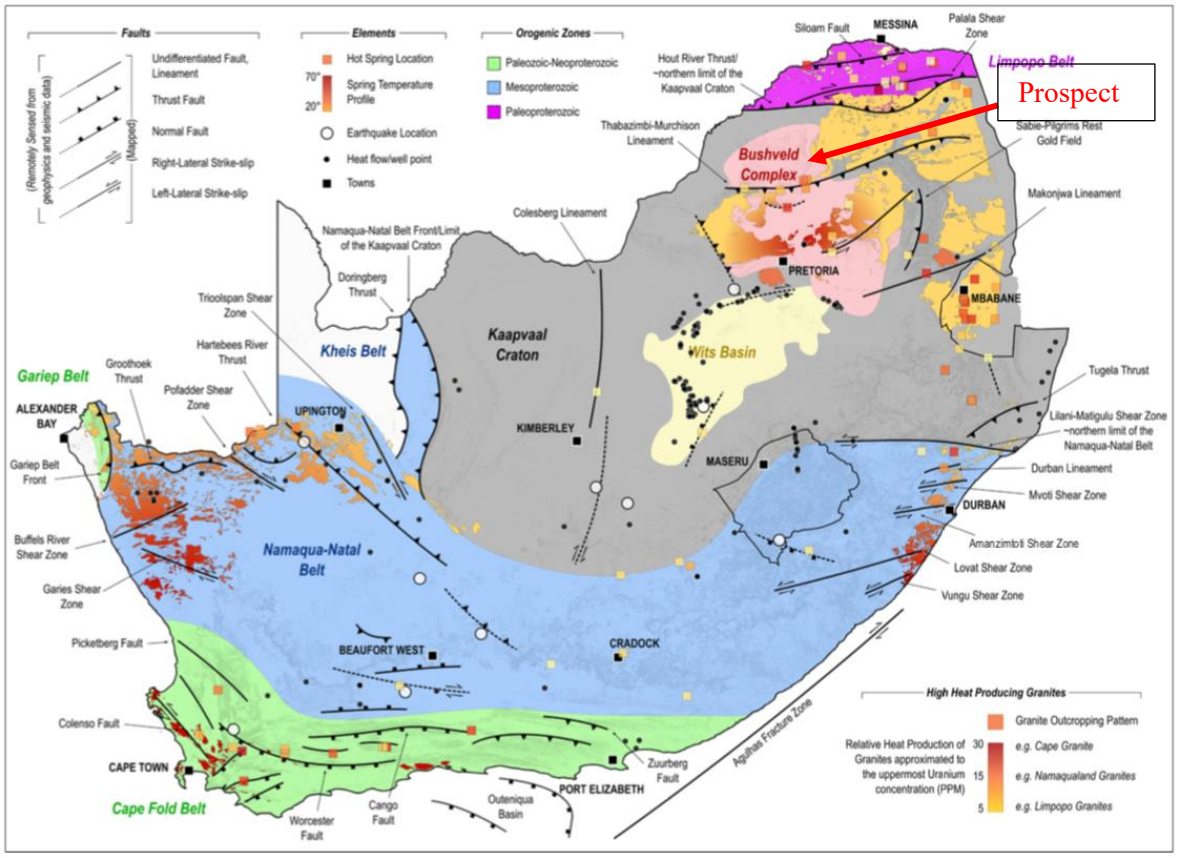


Fig. 4: Location map of the Bushveld Complex and the mine (Dhansay 2017).

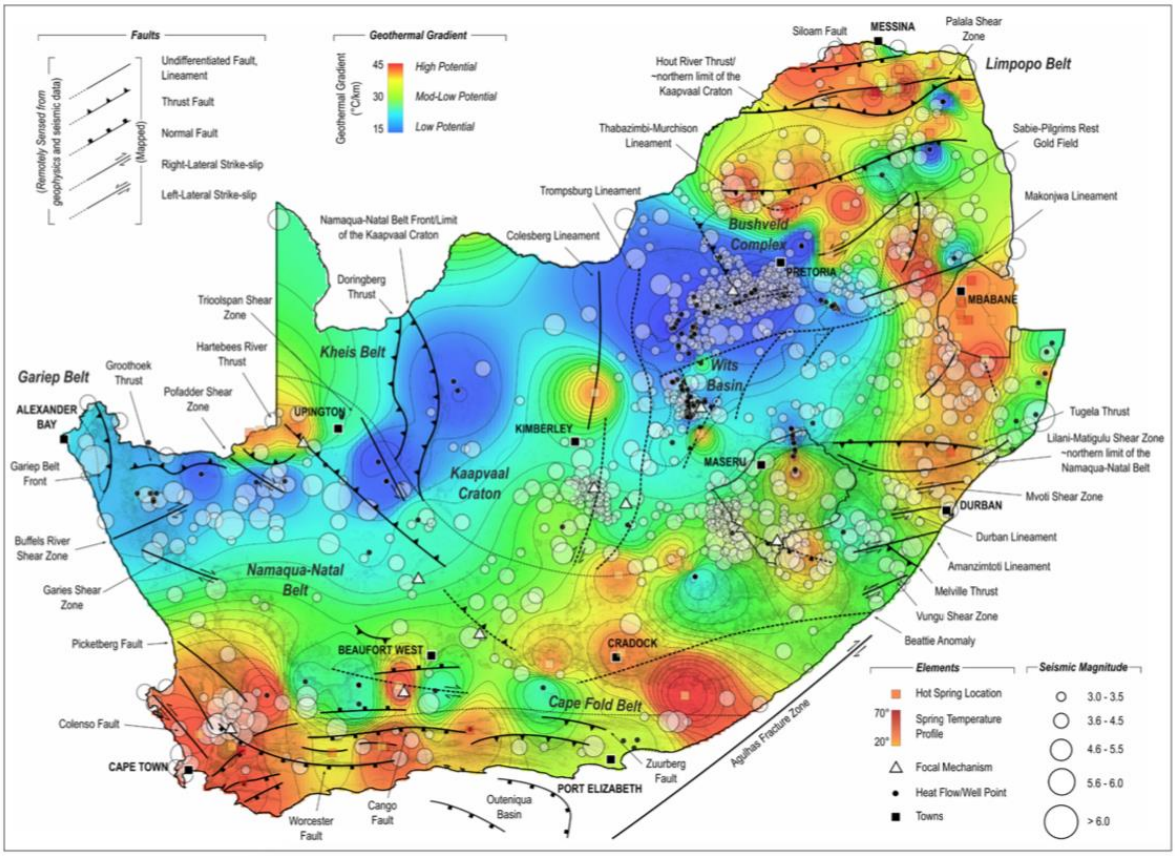


Fig. 5: Map of geothermal gradient (Dhansay 2017).